

BINOMIAL PRACTISE QUESTIONS

1. There are 10 members on a committee. The probability of any member attending a randomly chosen meeting is 0.9. The committee cannot do business if more than 3 members are absent. What is the probability that 7 or more members will be present on a given date?
2. A salesman has a 20% probability of making a sale to any customer who enters his department. On a typical day, he will meet 30 customers. What minimum number of sales will he have an 88% certainty of making for any given day?
3. An unbiased coin is tossed 10 times. Find the probability that there are fewer than 3 heads tossed.
4. A farmer plants 12 saplings. On average, 15% of saplings planted fail to survive their first winter. Find the probability that more than one of his saplings will die in that first winter.
5. ACME Inc. produces super-tactile widgets for the latest DIY craze, and they are sold in boxes of 8. Unfortunately, 2% of them are faulty. Find the probability that a given box has at least one faulty widget in it.
6. At Covington High School, 20% of sixth-form mathematics students retake their C1 module. There are 15 students in Mr Smither's class; find the probability that precisely four of them retake this module.
7. Brian enjoys his pizza. So much so that he eats pizza every night. He likes both pepperoni and mushroom, and each night he chooses randomly and independently between these two options. Given that each night he picks the mushroom option with probability 0.35, find the probability that he eats between 2 and 4 (inclusive) mushroom pizzas this week (7 days).
8. The probability of rain on any given day in June in Cambridge is 0.8. Assuming that the weather on each day is independent of the weather on other days, find the probability that it rains on at least 25 days in June.
9. There were ten green bottles sitting on the wall. The probability of a green bottle accidentally falling is 0.95. What is the probability that fewer than 8 of the green bottles accidentally fall?
10. A packet of vegetable seeds has a germination rate of 96%. What is the probability that exactly 10 of 12 seeds planted will sprout?